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Engineering Innovation Challenge 2023

Significant - Sign Language App for Social Inclusivity & Well-Being

Team U-17

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Project Abstract

In 2022, approximately 1% of people worldwide rely on sign languages as their primary means of communication. This poses a challenge when interacting with the remaining 99% of the population who do not understand sign language. Unlike spoken languages, the straightforward solution of employing a translator does not apply. Besides, different parts of the world employ different systems of sign languages, widening the divide between the deaf community and the general public.

Communication difficulties often result in emotional stress and social isolation. Therefore, cultivating social inclusivity by bridging communication gaps is crucial to promoting personal well-being and becomes a key characteristic of a resilient city. As such, Team U-17 has developed a software application ***Significant*** (Sign-if-i-can't) to address this communication barrier by facilitating conversion from sign languages to text/audio. This solution aims to enable people to utilize speech-to-text conversion to generate text that can be read by individuals with hearing impairments. Conversely, the application captures and records the gestures of deaf individuals through the device's camera, employing a trained AI model to recognize these gestures. The recognized keywords are then processed using ChatGPT to construct comprehensive sentences. The resulting text and corresponding speech are displayed on the screen and played through the phone's speaker. To train the sign language model, open-source sign language datasets for ASL (American Sign Language) and common sign languages can be sourced online. To enhance learning efficiency, we propose the use of transfer learning in the training process.

Project Report

1. Background & Research, Literature Review

In 2022, there were 72 million sign language users worldwide (National Geographic, 2022). However, the overall understanding of sign language in the general population is limited. Besides, different countries employ distinct sign languages such as ASL, BSL and SgSL, which increases the difficulty to communicate with the deaf community.

The existing technology, the Hand Talk app can only translate words or speech into sign language, which does not allow two-way communication. Moreover, it can only support ASL, leaving out other sign languages.

2. Problem Statement

How can technologies be leveraged to enhance communication accessibility for the deaf community, promoting well-being and fostering a more inclusive society, contributing to resilient city living?

3. Solution Description

Significant is an app that can bridge the hearing-disabled and the general public:

1. Facilitate two-way communication:
    a. Translate sign language to spoken language in text/speech
    b. Convert speech to text
2. Encompass multiple sign languages and spoken languages

4. App Design & Interface

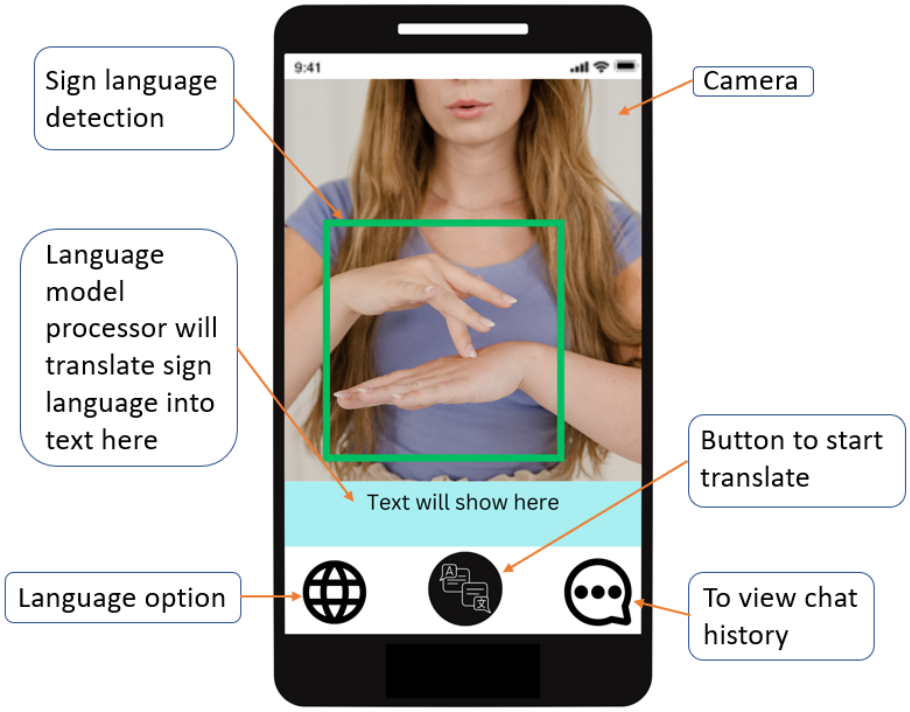


Figure 1. Main Interface

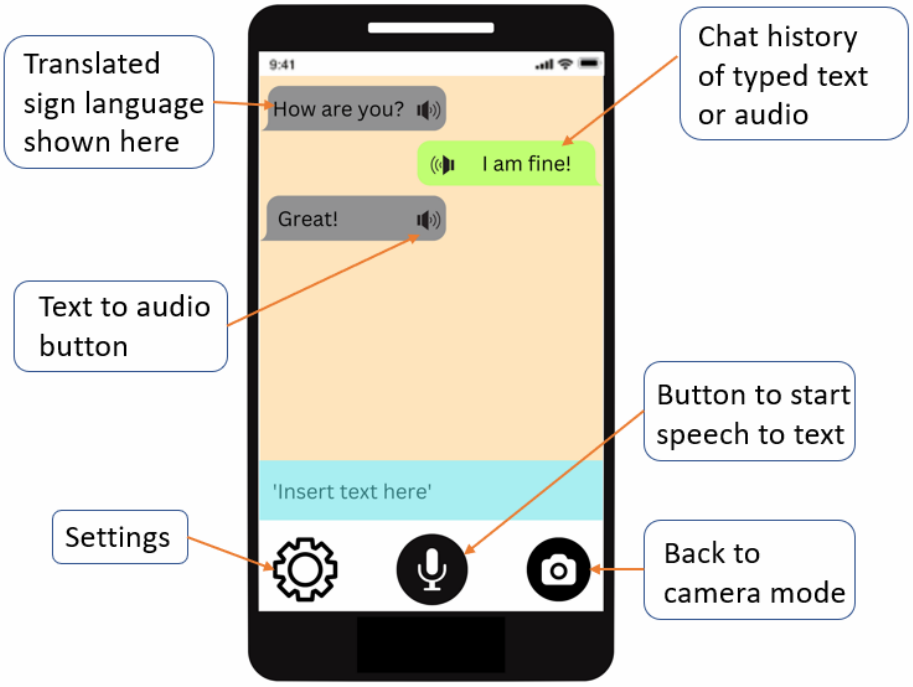


Figure 2. Dialogue Interface

5. Workflow of the App

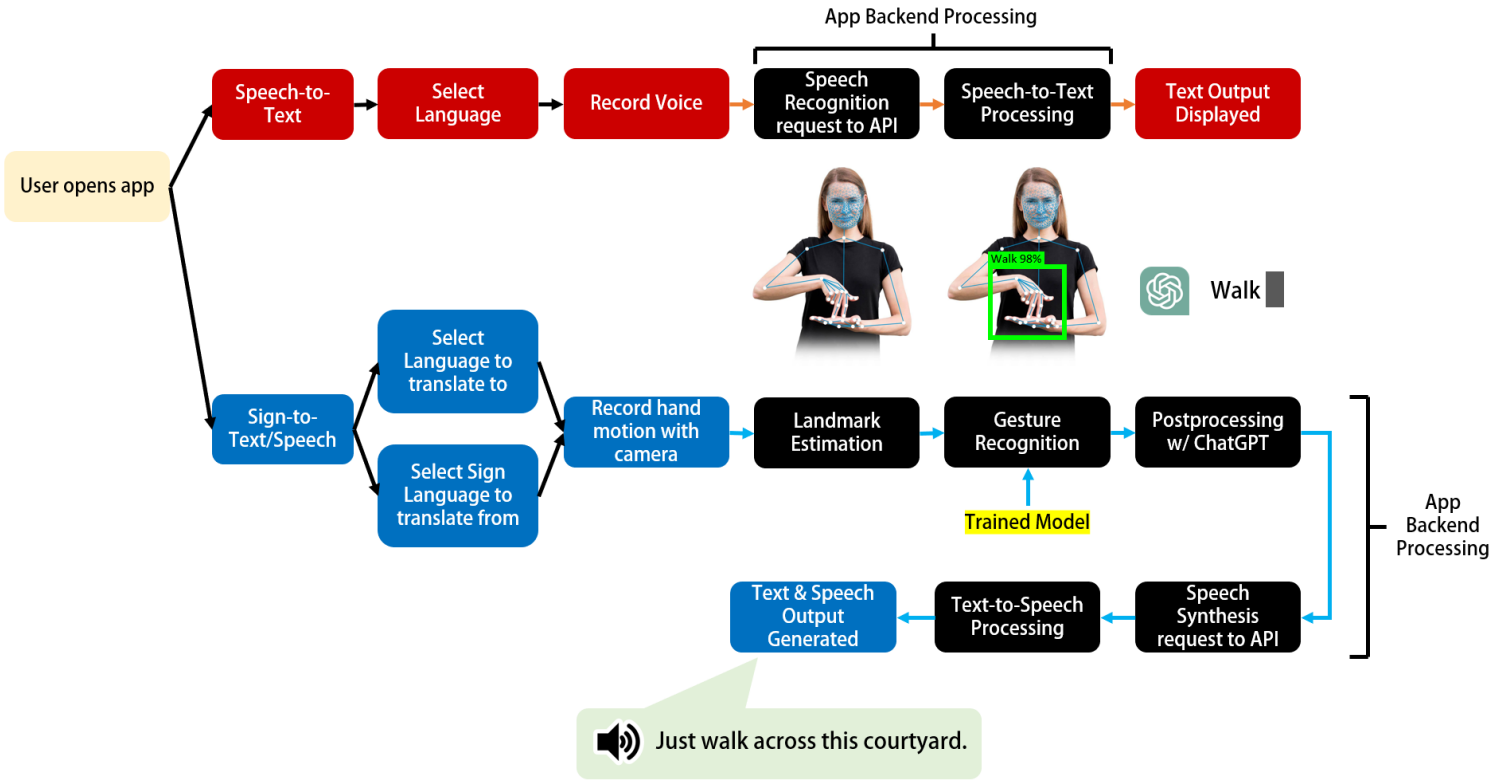


Figure 3. App Workflow

6. Data Analysis & Machine Learning

Google's MediaPipe (MediaPipe, n.d.) provides an efficient solution to machine learning and gesture recognition.

Open-source sign language datasets, and transfer learning (Kommineni, 2019) are proposed to train the sign language model (Refer to Appendix). MediaPipe's Model Maker can be utilized (MediaPipe, n.d.).

7. Risk & Safety

a. Privacy Concerns

To safeguard privacy, explicit user consent is acquired, limited data is collected, captured information is encrypted and privacy regulations are adhered to.

b. Accuracy and Reliability

To avoid information loss from language entropy, a language processing model (ChatGPT) is used to convert the grammar of the sign-to-spoken language.

8. Conclusion

Significant aims to remove all communication barriers and reduce social isolation for the deaf community worldwide. This will be a huge step for humanity in fostering resilient cities that embrace diversity and inclusivity.

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Appendix

Open-source sign language datasets such as the ASL dataset on Kaggle (Kaggle, 2017) and the sign language recognition datasets consolidated on GitHub (Quiroga, 2022) can be found.

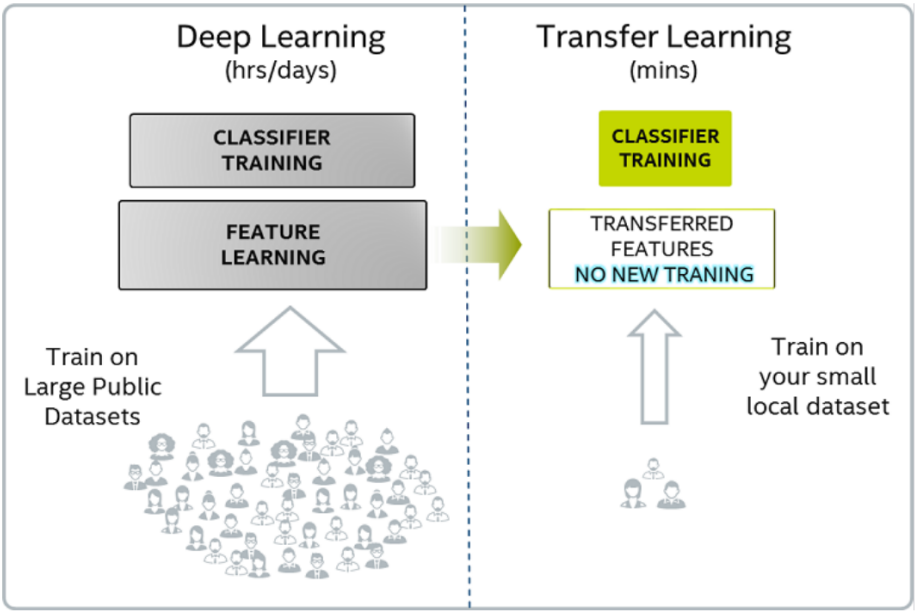


Figure A1. Transfer learning (Kommineni, 2019)

Transfer learning is where previously learned knowledge by a model is applied to another task. This can increase the efficiency of the task and reduce the size of the dataset needed as sign language includes image, video, and natural language processing which require a high amount of resources.